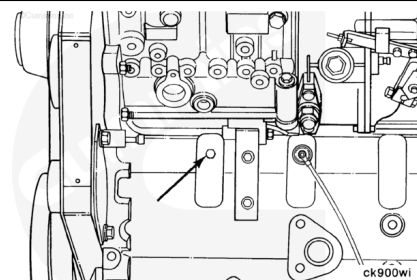


Trainings ▾

Measure

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Locate the appropriate port for measuring oil pressure. On most engines a port can be found along the main oil gallery. If this port is **not** accessible, another port can be located on the top of the oil filter head.

Remove the plug.

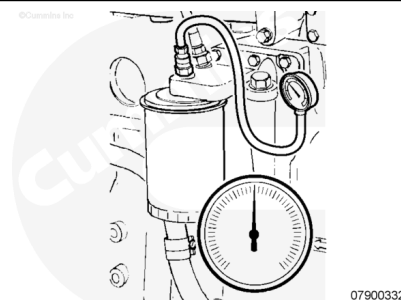
Note : It is preferred that oil pressure be checked at the main oil gallery, if accessible.

Install a Compuchek® fitting.

⚠ CAUTION ⚠

If the lubricating oil pressure does not develop within 15 seconds, shut down the engine to reduce the possibility of internal damage.

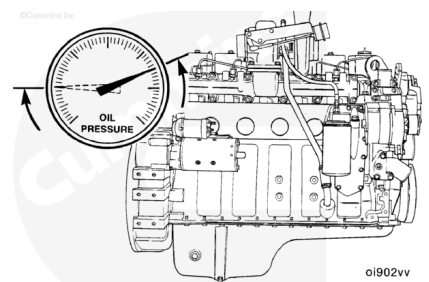
Connect a pressure gauge. Start the engine.



Allow the engine to run and achieve operating temperature.
Check for leaks.

Record the lubricating oil pressure readings at idle.

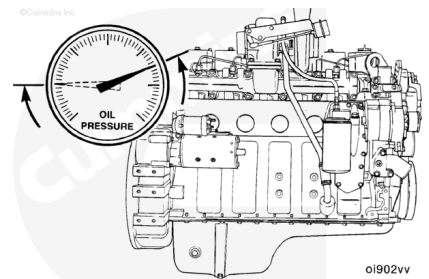
kpa		psi
69	MIN	10



Increase engine speed to rated speed and hold for 30 seconds.

Record the lubricating oil pressure readings at rated engine speed.

kpa		psi
207	MIN	30

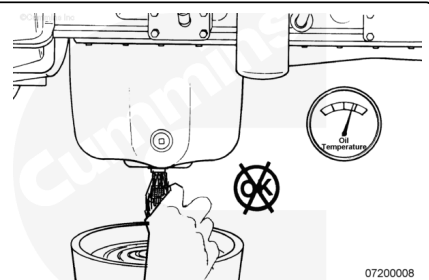


Remove the oil pressure gauge/Compuchek® fitting and install the previously removed plug. For pipe plugs, refer to Procedure 017-007 (/qs3/pubsys2/xml/en/procedures/40/40-017-007.html). For straight thread plugs, refer to Procedure 017-011 (/qs3/pubsys2/xml/en/procedures/100/100-017-011.html).

Drain

⚠ WARNING ⚠

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⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Note : For most engines use a container that can hold at least 20 liters [21 qt] of lubricating oil. Some engines may be equipped with an increased capacity oil pan

requiring a container that can hold 28 liters [30 qt] of lubricating oil.

Operate the engine until the coolant temperature reaches 60°C [140°F].

Shut off the engine.

Remove the oil drain plug. Drain the oil immediately to be sure all the oil and suspended contaminants are removed from the engine.

If performing an oil drain as part of a service maintenance interval, remove and replace the oil filter. Refer to Procedure 007-013 (/qs3/pubsys2/xml/en/procedures/40/40-007-013-tr.html).

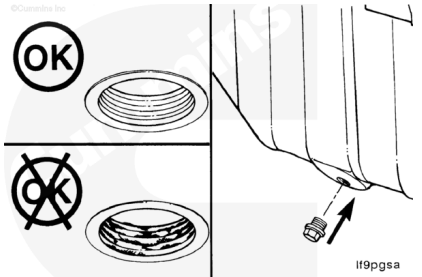
Fill

Clean and check the lubricating oil drain plug threads and sealing surface. Use new sealing washer, if damaged.

Install the lubricating oil pan drain plug.

	n.m		ft-lb
M18	60	MIN	44
M22	80	MIN	59

	n.m		ft-lb
M22	60	MIN	44



Note : Use a high-quality 15W-40 multiviscosity oil, such as Cummins Premium Blue®, or its equivalent, in Cummins® engines. Choose the correct oil for your operating climate. Refer to Cummins Engine Oil Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html)



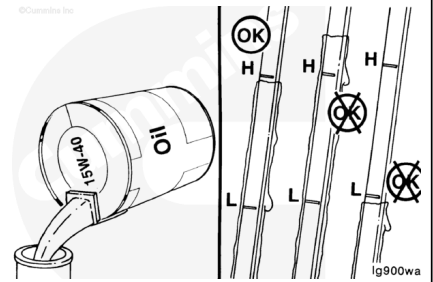
Fill the lubricating oil pan with clean 15W-40 lubricating oil to the low level. Refer to Procedure 018-017 (/qs3/pubsys2/xml/en/procedures/40/40-018-017-tr.html)

(Lubricating Oil System in Section V) for common lubricating oil pan capacities, or by looking up the oil pan option for the engine serial number on QuickServe OnLine.

Note : When filling the oil pan, use the fill tube on the side of the engine rather than on top of the rocker lever cover.

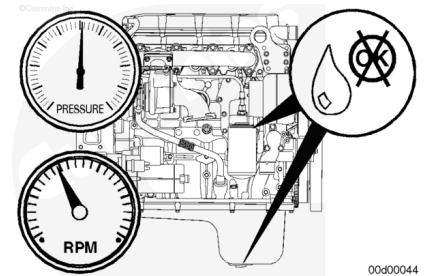
Note : If the engine is not equipped with a side-fill, wait at least 5 minutes before measuring the oil level with the dipstick to allow the oil to drain to the pan.

Service Tip: If the oil capacity of the oil pan is **not** known, fill the lubricating oil pan to the smallest oil pan capacity listed in Procedure 018-017 (/qs3/pubsys2/xml/en/procedures/40/40-018-017-tr.html) (Lubricating Oil System in Section V) for the engine being worked on. Then add 1 quart of oil at a time until it reaches the high mark on the dipstick. Record the number of quarts added so that capacity is known the next time the oil is drained.



Idle the engine to inspect for leaks at the drain plug and, if replaced, the oil filter seal.

Note : Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm that the correct oil level is in the oil pan.



Shut off the engine. Wait approximately 5 minutes to let the oil drain from the upper parts of the engine. Check the level again.

Add oil as necessary to bring the oil level to the H (high) mark on the dipstick.

